Snowflake

Real-Time Fraud Detection — Business Case

Prepared for Zilch • Head of Fraud • Executive Leadership

Confidential
Customer Engineering
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THE PROBLEM ZILCH IS TRYING TO SOLVE

Fraudsters test stolen cards in rapid bursts — often 5–10 low-value transactions within seconds. If your fraud model only sees **stale features**, every transaction in a burst looks clean until it's too late. The loss isn't just the fraudulent spend: it's chargebacks, issuer fees, reputational risk with Thredd, and the engineering cost of building the detection capability from scratch.

WHAT SNOWFLAKE DELIVERS — PROVEN IN TECHNICAL VALIDATION

<2s	~14ms	1	Zero
Feature freshness out of the box	End-to-end scoring per transaction	Platform replacing 7+ managed services	Custom ETL pipelines built or maintained

CURRENT COMPLEXITY VS. SNOWFLAKE APPROACH

WHAT ZILCH HAS TODAY

- ✓ Working fraud scoring endpoint — XGBoost model served today via Kinesis → SageMaker → DynamoDB → Triton pipeline
- ✓ Transaction data already in Snowflake — the data foundation is in place; the gap is specifically the real-time feature pipeline sitting between Snowflake and the model
- ⚠ Feature freshness not yet built — burst detection window is undefined; card-testing transactions look clean until the pipeline catches up
- ⚠ Training (SageMaker/RAPIDS) and serving (Triton/DynamoDB) are separate stacks — feature drift risk on every retrain without careful alignment
- ⚠ 10 services across 3 AWS service families — each scales, fails, and deploys independently; incidents require multiple teams to isolate root cause

WHERE SNOWFLAKE CLOSES THE GAP

- ✓ Fills the specific gap — sub-2s freshness with no custom build, covering the card-testing detection window the current stack is missing
- ✓ Unified feature logic in training and serving — structurally eliminates the drift risk between your SageMaker and Triton paths
- ✓ One platform to operate at scale — Snowpipe Streaming, Online Feature Store, SPCS, and Model Registry replace 7+ independently managed services
- △ Online Feature Store is in Public Preview — not yet GA; Snowflake provides enterprise support terms and committed SLAs for POC engagements
- △ No data migration required — your data already lives in Snowflake; the AWS pipeline under construction can be replaced rather than ported. Snowflake Customer Engineering can accelerate onboarding

PLATFORM SIMPLIFICATION

CURRENT AWS/NVIDIA STACK — 7+ SERVICES

Amazon Kinesis • AWS Lambda • Amazon S3 • AWS Glue ETL • RAPIDS Preprocessing • Amazon SageMaker • NVIDIA Triton • fraud-engine-service • Amazon EKS • Amazon DynamoDB

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SNOWFLAKE — ONE PLATFORM

Snowpipe Streaming • Online Feature Store • Snowpark ML Training • Model Registry • SPCS Scoring Service

BUSINESS IMPACT — ESTIMATED COST SAVINGS

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Faster Detection = Less Fraud Loss

When fraudsters test a stolen card, they fire 5–10 rapid transactions within seconds. Without real-time data, the fraud model can't see the burst forming — every transaction looks clean until it's too late. With Snowflake, **the pattern is visible from the second transaction** and blocked before the fraud completes.

£660K–£780K / year
→ £1.3M–£1.6M at 2× volume
24M annual transactions × 0.05% fraud rate = 12,000 incidents/yr × £500 avg cost = £6M total exposure. Card-testing attacks estimated at 20% of fraud volume (£1.2M). Real-time detection catches ~55% more burst events vs. a model on stale data.

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Engineering Cost Reduction

The 10-service stack requires ~1.75 FTE-equivalents of ongoing maintenance. Snowflake reduces that to ~0.5 FTE. AWS infrastructure saving is modest at current volume — the primary driver is headcount.

£170K–£220K / year
→ £220K–£300K at 3× volume (infra grows, Snowflake scales gradually)
~1.25 FTE freed at £130K fully loaded (primary driver). AWS infra saving £7K–£14K/yr at current scale — validated from unit pricing across Kinesis, Glue, EKS, SageMaker, DynamoDB.

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Faster Model Iteration

Fraud patterns evolve constantly — today, updating the model takes months, meaning new attack patterns go undetected while the build is in flight. Snowflake compresses that cycle to weeks, giving the fraud team **8 extra attempts per year** to improve accuracy and reduce both missed fraud and wrongly declined legitimate customers.

£200K–£300K / year
→ £400K–£600K at 2× volume
8 extra model updates/yr vs. quarterly cycles. Each update that catches a new attack pattern 4 weeks earlier saves ~£25K (3–4 patterns/yr). Each improvement in detection accuracy compounds across 24M annual transactions. Revenue upside from fewer false declines excluded.

WHY ACT NOW

Zilch's feature freshness component is still being built. **Every week it remains under construction is a week the fraud model runs blind to intra-session patterns.** Snowflake's Online Feature Store is in public preview today — the technical validation in this document was run on live infrastructure. Zilch can have sub-2-second freshness without waiting for a custom build to ship, and without the ongoing burden of maintaining it.

SUGGESTED NEXT STEPS

1. **POC on Zilch's Live Data**
   Run the validated pipeline on a sample of real transaction history. Measure freshness improvement against current baseline.

2. **Business Value Assessment**
   Using Zilch's actual fraud rates, transaction volumes, and engineering costs, we conduct a formalised TCO analysis — turning the estimates in this document into a business case grounded in your numbers.